ELEC 872 Week 5 Lab

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1 Experimental Setup and Reference Model

All experiments use the OPPORTUNITY dataset with the gesture task. The input is 113 channels from body-worn accelerometers, gyroscopes and magnetometers sampled at 30 Hz, and each window is labelled as one of 17 gestures or the Null class (no gesture). Training uses all five daily-living runs and the drill run of subject 1, and daily-living runs 1 and 2 and the drill run of subjects 2 and 3. Validation uses daily-living run 3 of subjects 2 and 3, and testing uses daily-living runs 4 and 5 of the same two subjects. Windows are 24 samples long (0.8 s) and move by 12 samples, which gives 41,399 training, 5,076 validation and 9,890 test windows. Of the test windows, 8,235 (83%) are Null.

The reference model follows the paper: four convolutional layers with 64 filters of length 5, two LSTM layers with 128 units, and dropout of 0.5, for 3,965,778 parameters. It is trained with RMSProp (learning rate 0.001, decay 0.9) on batches of 100 windows for up to 100 epochs. Training stops when the validation weighted F1 has not improved for 20 epochs, and the epoch with the best validation weighted F1 is evaluated once on the test windows.

We report two scores. Weighted F1 averages the per-class F1 with weights proportional to the number of test windows of each class, so the Null class dominates it. Macro F1 is the plain average over the 18 classes, so every gesture counts equally. Each configuration in Sections 1, 5, 6 and 7 was trained three times with different random seeds, and we report the mean and standard deviation over the three runs. These runs used one slice (1g.10gb) of an NVIDIA H100 GPU and took 6 to 13 minutes each.

The reference model reaches a test weighted F1 of 0.891 ± 0.004, a macro F1 of 0.580 ± 0.029 and an accuracy of 0.893. The paper reports a weighted F1 of 0.915 for the same task, so our reproduction is 0.024 lower. The single Colab run used as the baseline in the LSTM section gave 0.886, slightly below the lowest of our three runs (0.887). The macro F1 varies by 0.029 between seeds of the same configuration. In the following sections we therefore do not treat a difference smaller than about 0.03 in macro F1 as an effect of the change being tested.

2 Influence of LSTM on Model Performance

The paper's central claim is that adding recurrent layers on top of the convolutions improves recognition, because the LSTM models how the features unfold over time. It supports this with one comparison, DeepConvLSTM against a CNN with dense layers in place of the LSTMs. To test the paper's claim that the LSTM is necessary to capture the temporal dimension of the signal data, we train and validate two variations of the model:

- A no RNN version where the LSTM is replaced with two dense layers
- A GRU variant in which LSTM cells are replaced with GRU

	Accuracy	F1	Avg Precision	Avg Recall
Baseline	0.890	0.886	0.887	0.890
No RNN	0.874	0.857	0.860	0.874
GRU-Variant	0.896	0.896	0.901	0.896

Table 1: Baseline, no-RNN and GRU variants, one run each. Precision, recall and F1 are weighted by the number of test windows of each class, so the average recall equals the accuracy.

Table 1 shows that replacing the LSTM layers with dense layers lowered the weighted F1 from 0.886 to 0.857 and the accuracy from 0.890 to 0.874. The drop of 0.029 in weighted F1 is about seven times the seed-to-seed standard deviation of the reference model (0.004, Section 1), although each variant here was trained only once. The GRU variant reached a weighted F1 of 0.896, which is 0.010 above this baseline run. However, the baseline run is itself below the three-seed mean of the reference model (0.891), and against that mean the gain of the GRU shrinks to 0.005, about one standard deviation.

The results support the paper's claim only in part. Recurrent layers help, but the benefit is small: the model with dense layers still reaches an accuracy of 0.874 and a weighted F1 of 0.857, and these weighted scores are dominated by the Null class (83% of the test windows), so they may understate the difference on the gestures. The GRU does at least as well as the LSTM with fewer parameters per unit, so the gain seems to come from having a recurrent layer, not from the LSTM cell in particular. The windows are also only 24 samples (0.8 s) long, which gives the recurrent layers a short sequence to model. Section 4 looks at other window lengths.

3 Influence of CNN on Model's Performance

The paper argues that the convolutions learn useful features directly from the raw signals, so that no hand-crafted features are needed. Its Figure 8 shows the weighted F1 rising with the number of convolutional layers on the gestures with Null. It never tests a model without convolutions, and it does not vary the kernel

size or the number of filters. To test the CNN's influence on model success, the number of convolution layers was varied, and performance measured:

	Accuracy	F1	Avg Precision	Avg Recall
Baseline	0.890	0.886	0.887	0.890
Number CNNs = 0	0.854	0.806	0.789	0.854
Number CNNs = 2	0.893	0.893	0.896	0.893

Table 2: Effect of the number of convolutional layers, one run each. The baseline has four convolutional layers. The scores are weighted as in Table 1.

Table 2 shows that removing all convolutional layers lowered the weighted F1 from 0.886 to 0.806 and the accuracy from 0.890 to 0.854. The accuracy is only 0.021 above the 0.833 that a model predicting Null for every window would reach on this test set, and such a model would have a weighted F1 of about 0.757. The weighted precision of 0.789 is also far below the accuracy, whereas for the baseline the two differ by 0.003, which suggests that the model without convolutions handles the gesture classes poorly. With two convolutional layers the weighted F1 was 0.893, which is 0.087 above the model without convolutions and 0.007 above the four-layer baseline. The second difference is within the seed-to-seed variation of the reference model (standard deviation 0.004, and the baseline run is 0.005 below its three-seed mean of 0.891).

Most of the gain therefore comes from the first two layers, and we do not see a further rise from two to four layers as in the paper's Figure 8. With only three depths and one run each, we cannot say whether a single layer would already be enough. The result shows that learned convolutional features add a great deal to a recurrent model that is fed the raw signals. It does not show that hand-crafted features are unnecessary, because we did not compare against them. Removing the convolutions also changes the input of the LSTM from learned feature maps to raw channels, so the comparison mixes the loss of learned features with a change in model size. The kernel size and the number of filters were not varied here and remain untested.

4 How long should the window be?

The window decides how much of the past the model sees before it predicts. The reference model uses windows of 24 samples (0.8 s) that move by 12 samples (Section 1). The paper tried windows of 400, 500, 1400 and 2750 ms and reported the F1 of each gesture in its Figure 7. It found little change for most gestures, but the short drawer gestures did better when they fit inside the window. We repeat this comparison to test whether the weighted F1 depends on the window length, and whether the drawer gestures gain from longer windows in the same way.

We tested two window lengths, 0.67 s and 2.0 s.

Table 3 shows that the 0.67 s window reached a weighted F1 of 0.895 and the 2.0 s window 0.891, against 0.886 for the 0.8 s baseline run. The three values lie

	Accuracy	F1	Avg Precision	Avg Recall
Baseline (0.8s)	0.890	0.886	0.887	0.890
Window Size = 0.67s	0.897	0.895	0.899	0.897
Window Size = 2.0s	0.894	0.891	0.895	0.894

Table 3: Effect of the window length, one run each. The scores are weighted as in Table 1.

within 0.009 of each other, and the baseline run is itself 0.005 below the three-seed mean of the reference model (0.891). Against that mean the differences are +0.004 and 0.000, which is no more than the seed-to-seed standard deviation of 0.004 (Section 1). The accuracy shows the same pattern (0.897, 0.890 and 0.894 for 0.67, 0.8 and 2.0 s). With one run per window we therefore find no effect of the window length on the weighted F1, which agrees with the paper's finding that the window length changes little for most gestures. The 2.0 s window also gives the recurrent layers a sequence 2.5 times longer than the baseline without improving the weighted F1, so these results give no evidence that the LSTM benefits from more context. The comparison is only approximate, because a different window length also changes the windows the model is tested on.

The weighted F1 is dominated by the Null class, so it cannot show whether individual gestures changed. The paper reported that the short drawer gestures did better when they fit inside the window, so Table 4 gives the F1 of the six drawer gestures for the three windows.

Gesture	Baseline (0.8 s)	0.67 s	2.0 s
Open Drawer 1	0.366	0.437	0.485
Close Drawer 1	0.247	0.388	0.468
Open Drawer 2	0.317	0.379	0.394
Close Drawer 2	0.179	0.429	0.480
Open Drawer 3	0.564	0.676	0.571
Close Drawer 3	0.451	0.662	0.586
Mean	0.354	0.495	0.497

Table 4: Test F1 of the drawer gestures for each window length, one run each. Each gesture has 26 to 67 test windows, and the counts for the 0.67 s window differ by at most 3 from those of the other two windows.

Both new windows scored higher than the baseline run on all six gestures, and the mean F1 rose from 0.354 to 0.495 (0.67 s) and 0.497 (2.0 s). This is a much larger change than in the weighted F1, because the six drawer gestures make up only 273 of the 9,890 test windows (3%). The baseline run was also weak on these gestures. Its F1 of 0.247 and 0.179 for Close Drawer 1 and 2 is below the range of 0.31 to 0.57 that the drawer gestures reached in Section 7, so part of the gain is probably due to a low baseline run. The two new windows differ from each other by gesture and not on average. The 2.0 s window gave

a higher F1 for the four Drawer 1 and 2 gestures (by 0.015 to 0.080), whereas the 0.67 s window gave a higher F1 for the two Drawer 3 gestures (by 0.105 and 0.076).

These results do not clearly confirm the paper's finding. A gain from fitting inside the window would favour the longer window, but the shorter window also beat the baseline, and the 2.0 s window helped only some of the gestures. With one run per window, we do not interpret most of these differences, since a change of loss alone moved individual drawer gestures by 0.04 to 0.07 in Section 7 with no overall effect. In summary, window lengths between 0.67 and 2.0 s have no measurable effect on the weighted F1, and their effect on individual drawer gestures is unclear.

5 How Much Regularization Is Needed?

The paper uses dropout of 0.5 and no other regularization, and states in Section 5.4 that higher dropout made no difference, without giving numbers. We test whether dropout, weight decay and data augmentation change the test results.

Because the reference model already has no weight decay and no augmentation, removing either from it would leave it unchanged. We therefore switched on all three regularizers and removed one at a time. Dropout of 0.5 is applied to the convolutional features, between the two LSTM layers and before the output layer. Weight decay adds an L2 penalty of 10^{-4} to the RMSProp update. Augmentation multiplies every channel of every training batch by a random gain drawn from $\mathcal{N}(1, 0.1^2)$ and adds Gaussian noise with a standard deviation of 0.01 (the data are normalized to $[0, 1]$). To measure over-fitting, we also scored each saved model on the training windows with dropout and augmentation switched off, and compared its macro F1 there with the test macro F1.

	Test		Training	
Setting	Weighted F1	Macro F1	macro F1	Gap
Reference (dropout only)	0.891 ± 0.004	0.580 ± 0.029	0.902	0.32
All three regularizers	0.883 ± 0.007	0.562 ± 0.012	0.877	0.32
No dropout	0.885 ± 0.004	0.569 ± 0.013	0.940	0.37
No weight decay	0.881 ± 0.005	0.553 ± 0.041	0.852	0.30
No augmentation	0.890 ± 0.006	0.578 ± 0.016	0.919	0.34

Table 5: Regularization settings, mean ± standard deviation over three seeds. All three regularizers means dropout, weight decay and augmentation together; the last three rows each remove one of them. Gap is the training macro F1 minus the test macro F1.

Table 5 shows that the test macro F1 of all five settings lies between 0.553 and 0.580. The largest difference, 0.027, is smaller than the seed-to-seed variation

of the reference model (0.029). Adding weight decay and augmentation on top of dropout did not improve the results: the weighted F1 was 0.883 against 0.891 for dropout alone. Removing dropout raised the training macro F1 from 0.877 to 0.940 and widened the gap between training and test from 0.32 to 0.37 (Figure 1), while the test macro F1 stayed at 0.569 against 0.562.

Dropout therefore reduces over-fitting, but in this setting that does not change the test performance, and weight decay and augmentation have no measurable effect either. This agrees with the paper's statement that the regularization settings matter less than the architecture. With three seeds per setting, we cannot rule out effects smaller than about 0.03 in macro F1.

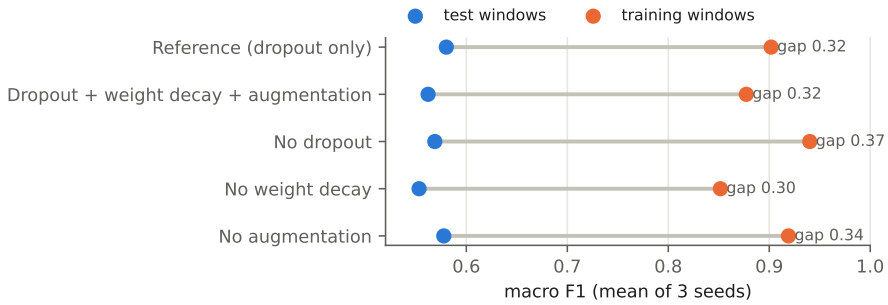


Figure 1: Macro F1 on the training windows (orange) and on the test windows (blue) for each regularization setting, mean of three seeds. The label gives the difference.

6 Does a Learning-Rate Warm-up Make Training More Stable?

The paper trains with RMSProp at a fixed learning rate of 0.001 and reports that it gave the best results, without giving numbers. We tested whether a learning-rate warm-up makes training more stable. The learning rate rises linearly from almost zero to 0.001 over the first 5 epochs and is updated after every mini-batch, which is 2,070 updates at 414 mini-batches per epoch. After the warm-up the training is identical to the reference.

We measured stability in three ways. The early loss variation is the standard deviation of each mini-batch loss relative to the mean of the surrounding 50 mini-batches, over epochs 1 to 5. The largest validation drop is the largest fall of the validation weighted F1 from one epoch to the next. The validation-loss increases count the epochs among the first 15 in which the validation loss went up.

With the warm-up, the early loss variation fell from 0.209 to 0.188 (Table 6), and the difference is larger than the spread between seeds. The largest valida-

	Reference	Warm-up (5 epochs)
Test weighted F1	0.891 ± 0.004	0.894 ± 0.004
Test macro F1	0.580 ± 0.029	0.604 ± 0.016
Early loss variation (epochs 1 to 5)	0.209 ± 0.004	0.188 ± 0.008
Largest validation weighted F1 drop	0.032 ± 0.019	0.021 ± 0.011
Validation-loss increases (first 15 epochs)	6.0 ± 1.0	6.7 ± 1.5
Best epoch	35 (21, 62, 21)	56 (36, 62, 71)

Table 6: Learning-rate warm-up, mean ± standard deviation over three seeds. Lower values of the three stability measures mean more stable training. The best epoch is the mean, with the three seeds in brackets.

tion drop fell from 0.032 to 0.021, but the seeds overlap, and the number of validation-loss increases did not change. Figure 2 shows one seed. During the warm-up the training loss falls more slowly, and the reference run shows a single sharp drop of the validation weighted F1 to 0.82 at epoch 38 that the warm-up run does not have. The warm-up also delayed convergence, as the best epoch moved from 35 to 56 on average.

The warm-up makes the first epochs somewhat smoother, but it does not give a clearly better model. The test macro F1 rose by 0.024 to 0.604, which is within the seed-to-seed variation of the reference. RMSProp at 0.001 already trains this model without serious instability.

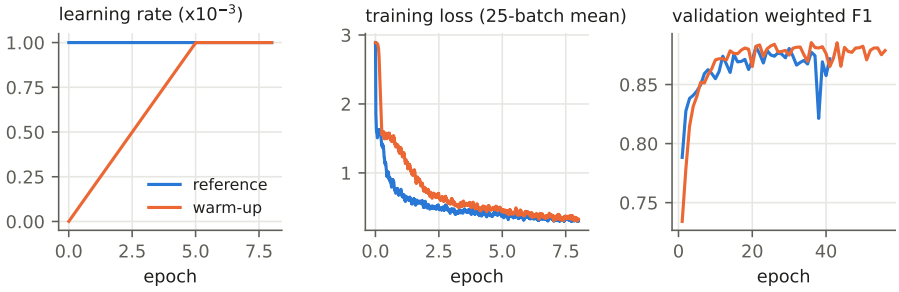


Figure 2: Learning rate (left) and training loss averaged over 25 mini-batches (middle) for the first 8 epochs, and validation weighted F1 over the whole run (right), for the reference (blue) and the warm-up (orange). One seed.

7 Does Handling Class Imbalance Help Rare Gestures?

The test set is dominated by the Null class, and several gestures have fewer than 100 test windows. The paper does not address this imbalance and reports only

weighted F1. We replaced the cross-entropy loss with the Focal Loss of Lin et al. (2017),

$$\mathrm{FL}(p_t) = -(1 - p_t)^{\gamma} \log p_t, \tag{1}$$

where p_t is the predicted probability of the true class. The factor $(1 - p_t)^{\gamma}$ reduces the loss of windows the model already classifies confidently, which in this dataset are mostly Null windows. We used $\gamma = 2$, the value Lin et al. found best, and no class weights, so only the shape of the loss changes. The best epoch is still chosen on the validation weighted F1.

	Paper	Cross-entropy	Focal Loss ($\gamma = 2$)
Test weighted F1	0.915	0.891 ± 0.004	0.892 ± 0.003
Test macro F1	not reported	0.580 ± 0.029	0.589 ± 0.029
Macro F1, 17 gestures	not reported	0.558	0.567
Null precision	not reported	0.942	0.947
Null recall	not reported	0.958	0.949

Table 7: Cross-entropy (the reference model) and Focal Loss, mean over three seeds ($\pm$ standard deviation where shown). The paper reports only the weighted F1.

The Focal Loss did not change the overall scores (Table 7). The weighted F1 was 0.892 against 0.891, and the macro F1 rose by 0.009, far less than the seed-to-seed variation of 0.029. Figure 3 shows the per-gesture F1. Nine of the 17 gestures improved, and over the 12 gestures with fewer than 100 test windows the mean change was 0.000. The drawer gestures, which are the rarest, moved in both directions: Close Drawer 1 and Open Drawer 2 improved by 0.07, while Close Drawer 2 and Open Drawer 1 dropped by 0.04. The Null recall fell from 0.958 to 0.949, which means the model predicted gestures slightly more often. Neither loss closes the gap of about 0.02 to the paper's weighted F1.

Focal Loss with $\gamma = 2$ alone does not help the rare gestures in this setting, and the drawer gestures remain between 0.31 and 0.57 F1. Focal Loss down-weights windows by how confidently they are classified, not by how rare their class is. Adding the class weights (α) of Lin et al. or sampling the rare gestures more often would be the next step to test.

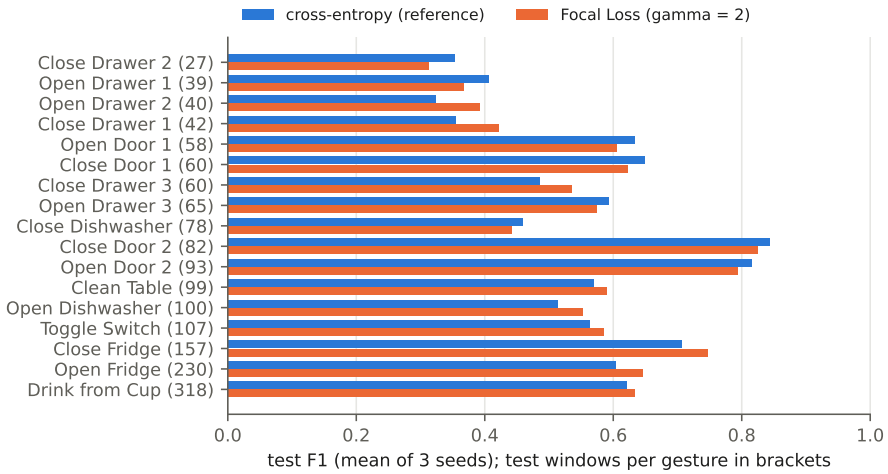


Figure 3: Test F1 of each gesture with cross-entropy (blue) and Focal Loss (orange), mean of three seeds, rarest gesture first. The number of test windows of each gesture is in brackets.